

Online Retail Sales Analysis - Power BI Dashboard

Project Objective:

The objective of this project is to analyze transaction data from an online retail store and build an interactive Power BI dashboard to track sales performance, identify top selling products and markets, and understand customer buying patterns. The project involved cleaning the raw data in Excel, then building a data model, DAX measures and a multi page report in Power BI.

Dataset Description:

The dataset used is the Online Retail dataset, which contains transaction records for a UK based online retailer selling gift items, from 1 December 2010 to 9 December 2011. The raw file had about 541,000 rows. After removing blank customer IDs, duplicate records and cancelled orders during the Excel cleaning stage, the final dataset used for the dashboard has 392,692 rows.

Column Description

Column Name	Description
InvoiceNo	Unique number given to each transaction. Numbers starting with C are cancelled orders.
StockCode	Unique code given to each product.
Description	Name of the product.
Quantity	Number of units of the product sold in that transaction.
InvoiceDate	Date the transaction was made.
UnitPrice	Price of one unit of the product, in GBP.
CustomerID	Unique number given to each customer.
Country	Country of the customer.
Sales	Quantity multiplied by Unit Price, calculated during cleaning.
Year	Year the transaction was made in, added for reporting.
Month Name	Month the transaction was made in, added for reporting.
Quarter	Quarter the transaction falls in, added for reporting.

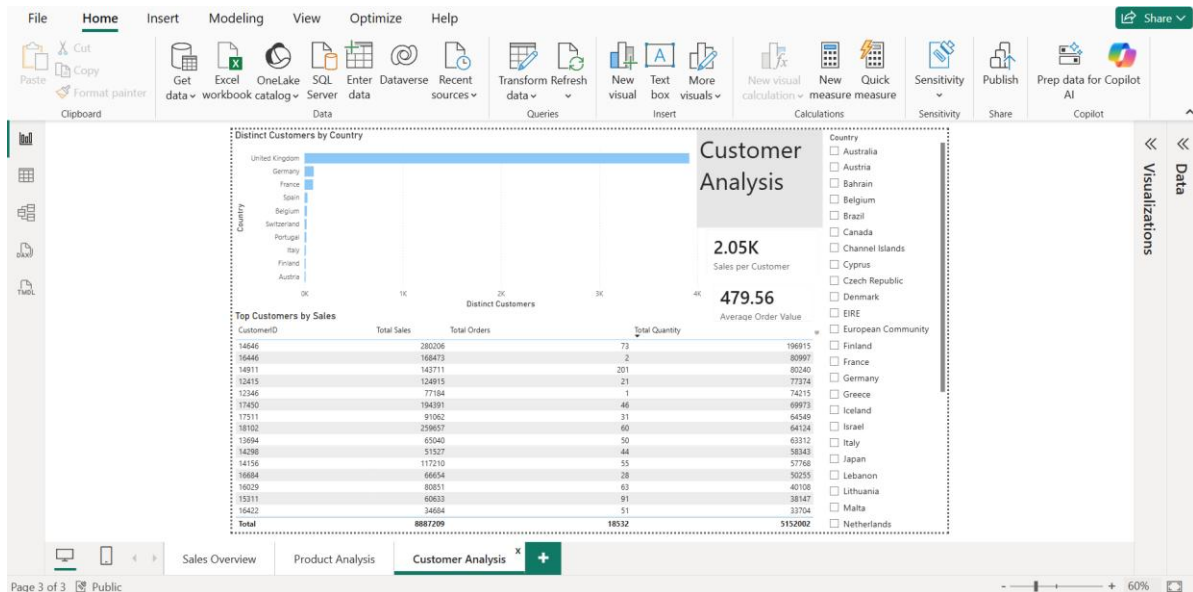
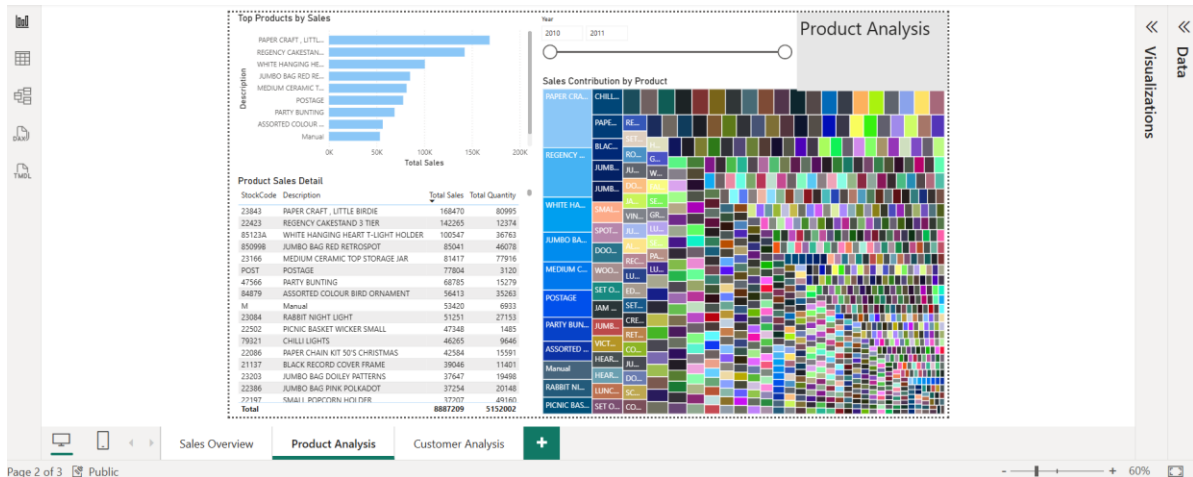
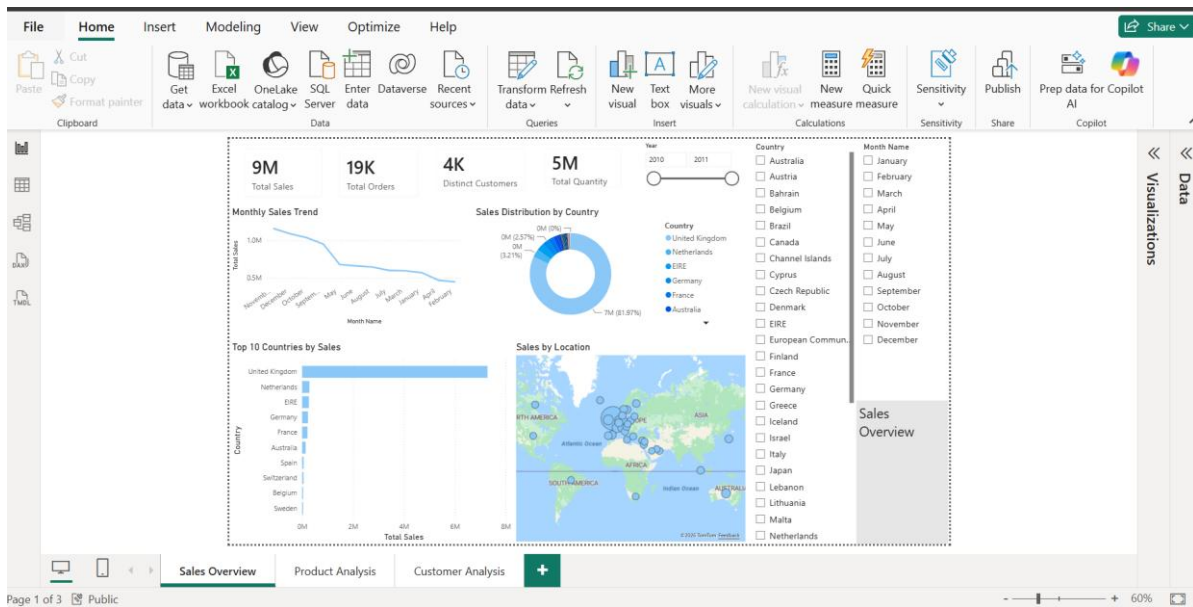
Data cleaning & transformation:

- Removed around 132,000 rows where Customer ID was blank, as these could not be tied to a customer.
- Removed duplicate rows.
- Removed cancelled orders (invoice numbers starting with C) and rows with negative quantity.
- Standardized the InvoiceDate column and added Year, Month Name and Quarter columns.
- Added a calculated Sales column (Quantity x Unit Price).
- Checked and corrected data types for Quantity, Unit Price, Customer ID and Sales.
- Built pivot tables in Excel to check totals before moving the data into Power BI.

Visualization and Insights

The cleaned data was loaded into Power BI and a star schema was built with a fact table (FactSales) and dimension tables for customer, product, country and date. Twelve DAX measures were created for the report, including Total Sales, Total Orders, Distinct Customers and Average Order Value. The final report has three pages - Sales Overview, Product Analysis and Customer Analysis, with slicers for year, month and country, and a drill down on the monthly sales trend chart.

Report/Dashboard screenshot:



- Descriptive analysis:
 - Total sales for the period were around 9 million pounds, from about 19,000 orders and 5 million units sold.

- Around 4,000 different customers placed orders, with an average order value of about 480 pounds.
- The United Kingdom made up about 82% of total sales, followed by the Netherlands, Ireland, Germany and France.
- Paper Craft, Little Birdie and Regency Cakestand 3 Tier were the two best selling products by value.
- Diagnostic Analysis:
 - Most of the sales come from the UK, which shows the business has not expanded much into other markets yet.
 - The sales trend drops off in the last month shown on the chart, but this is because the dataset ends partway through December 2011, not because of an actual drop in sales.
 - The best selling products are mostly low priced, high volume items, which is why quantity sold is high compared to total sales value.
 - A small group of customers account for a large share of total sales, so the business depends on a few high value, likely wholesale, buyers.
- Predictive Analysis:
 - This dashboard does not include forecasting, since the focus was on descriptive and diagnostic reporting.
 - Power BI's forecasting option could be added to the monthly sales trend chart later to estimate future sales.
- Prescriptive Analysis:
 - Focus on keeping the top spending customers, since a small group generates a large part of total revenue.
 - Look at ways to grow sales in countries like Germany, France and Australia, where sales are currently low.
 - Consider bundling popular low cost products with slower selling items to increase average order value.
 - Check whether the drop in the last month of the data is due to the data being incomplete or an actual decrease in orders.

Conclusion:

This project took a large raw sales dataset and turned it into a cleaned dataset in Excel and a working Power BI dashboard. The dashboard has three report pages with cards, charts, a map, slicers, drill down and bookmarks, and gives a clear view of sales performance, product performance and customer behaviour that can be used to support decisions on customer retention, market expansion and promotions.